

Disease Prediction: Smart Disease Prediction System using Random Forest Algorithm

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Abstract— People nowadays suffer from a variety of diseases as a result of their living habits and the state of the environment. As a result, predicting sickness at an early stage becomes a crucial task. A doctor's ability to establish accurate diagnosis solely on symptoms, on the other hand, is restricted. For the prevention and treatment of illness, an accurate and timely examination of any health-related problem is critical and challenging. In the case of a critical illness, the conventional method of diagnosis may not be adequate. There will be a huge requirement for Automated Disease Prediction System that will reduce these challenges. Developing a medical diagnosis system based on the Random Forest machine learning algorithm for disease prediction can aid in a more accurate diagnosis than the conventional way. The goal of constructing a classification system using a machine learning algorithm i.e Random Forest will substantially enable physicians in anticipating and detecting diseases at an early stage, greatly assisting in the resolution of health-related issues. For the analysis, a sample of 4920 patient records with 41 disorders was chosen. A total of 41 diseases made up the dependent variable. We enhanced 95 of the 132 independent variables (symptoms) that are closely related to illnesses. This paper illustrates a disease prediction system constructed using the Random Forest Machine Learning algorithm. Experiments were conducted with a standard symptoms dataset, and this model achieved 95 % classification accuracy. Machine learning and the Python programming language with the Tkinter Interface were used to create this disease prediction using Random Forest.

Keywords — random forest, disease prediction

I INTRODUCTION

Medicine and healthcare are important aspects of both the economy and human life. When specific data mining technologies are applied correctly, important information can be mined from vast databases, allowing medical practitioners to make more informed decisions and enhance health safekeeping. The goal is to help the doctor by using the classification. Malaria, dengue fever, impetigo, diabetes, migraine, jaundice, chickenpox, and other diseases and health problems have a significant impact on one's health and, if left untreated, can lead to death.

The healthcare business can make more informed decisions by "mining" its massive database or extracting hidden patterns and linkages. Between the world, we live in now and the world that existed just a few weeks ago, there has been a significant shift. Physicians and nurses are doing everything they can to save people's lives in this environment, even if it means putting their own lives in danger. There are also some isolated villages without medical services.

TABLE I: SYMPTOMS 1

Back Pain	Chest Pain	Scurrying
Constipation	Bloody stool	Passage of gases
Abdominal pain	Depression	Weakness in limbs
diarrhea	Irritation in anus	Fast heart rate
Mild fever	Neck pain	Internal itching
Yellow urine	Dizziness	Painful walking
Yellowing of eyes	Cramps	Prominent veins on calf
Acute liver failure	Bruising	Fluid overload
Fluid overload	Obesity	Excessive hunger
Swelling of stomach	Swollen legs	Blackheads
Swollen lymph nodes	Irritability	Pain during bowel movements
Malaise	Swollen blood vessels	Rusty sputum
Blurred and distorted vision	Muscle pain	Mucoid sputum
Phlegm	Pain in the anal region	Puff face and eyes
Throat irritation	Brittle nails	Hip joint pain
Redness of eyes	Belly pain	psoriasis
Sinus pressure	Enlarged thyroid	Family history
Runny nose	Slurred speech	Swollen extremities
Congestion	Knee pain	Spinning movements
Chest pain	Yellow crust ooze	Bladder discomfort
Toxic look	Loss of smell	Altered sensorium
Palpitations	Movement stiffness	Dyschomic patches

TABLE II SYMPTOMS 2

Lack of Concentration	Swelling joints	Visual Disturbances
Receiving unsterile injections	Stiff neck	Distension of abdomen
Puss filled pimples	Muscle weakness	Blood in sputum
Silver like dusting	Red sore around the nose	Small dents in nails
Blister	Watering from eyes	Unsteadiness
Drying and tingling lips	Weakness of one body side	The continuous feel of urine
Abnormal menstruation	Receiving blood transfusion	Stomach bleeding
Increases appetite	History of alcohol consumption	Inflammatory nails

TABLE III: DISEASES

Fungal infection	Malaria	Varicose Veins
Allergy	Chickenpox	Hypothyroidism
Gerd	Dengue	Vertigo
Chronic cholestasis	Peptic ulcer disease	Acne
Drug reaction	Hepatitis A	Urinary tract infection
Piles	Hepatitis B	Psoriasis
AIDS	Hepatitis C	Impetigo
Diabetes	Hepatitis D	Hyperthyroidism
Gastroenteritis	Hepatitis E	Hypoglycemia
Bronchial Asthma	Alcoholic hepatitis	Cervical Spondylosis
Hypertension	Tuberculosis	Arthritis
Migraine	Common cold	Osteoarthritis
Paralysis	Pneumonia	Typhoid
Jaundice	Heart attack	

Virtual doctors who prefer phone appointments and online suggestions with low cost. however, this is not possible in an emergency. Machines are always thought to be superior to people because, in the absence of human error, they can complete jobs more quickly and consistently.

A disease predictor, often known as a virtual doctor, may properly anticipate a patient's illness without requiring human intervention. According to a study, 70 percent of Indians suffer from the general disease, and 25 percent face death as a result of early ignorance. The lack of willingness to see a doctor, as well as a lack of time, is the most typical cause of ignorance. People have become so preoccupied with their daily lives that they have no time to schedule an appointment or contact a doctor, resulting in fatal sickness. Early disease detection and diagnosis may be aided by the model prediction. For this study, we used an online dataset to generate machine learning-based predictions on some test data. It develops a model to predict the disease based on the symptoms.

II RELATED WORK

Using machine learning algorithms, several studies have been conducted to forecast diseases based on symptoms displayed by a person. Monto *et al.* [19] created a statistical model that could predict whether or not a patient was infected with influenza. They included 3744 unvaccinated adults and adolescents who had a fever and at least two other symptoms of influenza. Of the influenza virus, A total of 2470 people out of 3744 were confirmed to have influenza as determined by the laboratory. Their model has a 79 percent accuracy rate based on this information. The random forest machine-learning technique was utilized by Sreevalli *et al.* [18] to predict disease based on symptoms. For disease prediction, the method required little time and money. The algorithm proved successful, with an accuracy of 84.2 percent.

Langbehn *et al.* [17] created some methods to help diagnose Alzheimer's disease. The least realistic model was RUSBoosted trees. Subspace, Medium, and Coarse KNN models all fared worse than Fine KNN models. Coarse KNN was the least efficient KNN model. Gaussian and Kernel Naive Bayes algorithms were comparable to KNN models in terms of accuracy, but they were less accurate. The accuracy of the model was 79 percent. With the use of SmoteBOOST and wRACOG algorithms, they constructed classification models to detect reliable absolute changes in scores. Different Machine Learning (ML) approaches are used in cancer disease prediction to get the result effectively and to produce accurate decision making [14].

In [9] Karayilan *et al.* developed a backpropagation algorithm for artificial neural networks that predicts heart disease. The neural network was fed 13 clinical features as input and then trained to predict the chance of getting heart disease with 95% accuracy using the backpropagation model. For the effective prediction of a chronic disease pandemic, Chen *et al.* [8] streamlined different machine learning algorithms. The information gathered for the training was insufficient. A latent component model i.e CNN-MDRP was utilized to solve this problem. The algorithm's accuracy rate was 94.8 percent. For monitoring 80 infectious diseases in six groups, Chae *et al.* [3] utilized four different deep learning models. All models tested performed better than DNN and LSTM when occurrences were large. However, LSTM produced near predictions when occurrences were small.

Haq *et al.* [4] used a database that contained data on patients suffering from any type of heart disease. Relief, minimum redundancy, and maximum relevance (mRMR), as well as least absolute shrinkage and selection operator, were

used to extract features, which were then cross-verified using the K-fold model. The collected attributes were sent into six different machine learning algorithms, which categorized them based on whether or not they had heart disease. Mohan *et al.* [2] created an excellent cardiac disease prediction system. Achieved an accuracy of 88.4 percent in the prediction model for heart disease using a hybrid random forest with a linear model (HRFLM). ML algorithms were used by Maniruzzaman *et al.* [1] to categorize diabetes. To identify diabetes illness risk variables, researchers employed logistic regression (LR). The ML-based system had an overall accuracy rating of 90.62 percent.

The author of the study "Disease Prediction System utilizing Data Mining Techniques" [12] discusses data mining techniques such as association rule mining, classification, and clustering to analyze various heart-related problems. The fields addressed in the paper "A study on data mining prediction strategies in the healthcare sector" [13] include the information Discovery method (KDD), which is a way of modifying low-level data into high-level data.

In the publication "Predicting Disease by Using Data Mining Based on Healthcare Information System" [16], the information mining approach is used to predict high blood pressure from patient medical records for 8 different diseases. The information was taken from a real-world healthcare system that included medical records. The study "An approach to devising an Interactive software solution for smart health prediction using data mining" [15] proposes to create a computerized system that can assess and maintain your health by recognizing symptoms. Alshurafa *et al.* [11] used baseline surveys to collect labeled patient data as well as real-time mobile data. It shows that the K-Nearest Neighbour algorithm is vulnerable to noisy data.

The heart disease prediction algorithm in [10] with symptoms collects data from patients via a questionnaire and feeds it into a Support Vector Machine Algorithm. Prabakaran *et al.* [7] show an online system that uses Naive Bayes to answer tough inquiries to diagnose heart illness. Used methodologies like Leave-One-Out and Cross-Validation. P for evaluating the accuracy of classification methods. Chiang *et al.* [6] used data from 8 persons over a month obtained using health monitoring devices such as Fitbit and Omron. RF feature selection was used to perform feature ranking. To eliminate missing data and outliers [5] used additive Least Square and Support Vector Machine.

III METHODOLOGY

For the analysis, a sample of 4920 patient records with 41 disorders and 132 symptoms was chosen. This study enhanced 95 of the 132 independent variables (symptoms) that are closely related to illnesses. Data preprocessing is a data mining approach that transforms or encodes raw data. Data cleaning and data reduction are the preprocessing strategies employed in this study. Random Forest is a supervised learning technique for classifying and forecasting data. It is, however, largely used to overcome categorization problems. The random forest technique, on the other hand, builds decision trees from data samples, extracts predictions from each, and then votes on the best alternative. You can learn how the Random Forest algorithm works by following the steps below.

Step 1: First, select random samples from a dataset.

Step 2: This algorithm will create different decision trees using two or more selected features with a replacement

Step 3: After the creation of all decision trees, the disease will of symptoms with the previous records collected from the be

predicted in each decision tree.

Step 4 Mode value will be calculated for each predicted a disease that refers to the voting process.

Step 5: The disease obtained with the highest mode value is treated as a final estimation of the disease.

Algorithm Used	Accuracy Score	Confusion matrix	
		Correctly classified	Incorrectly classified
Random Forest	0.95121	39	2

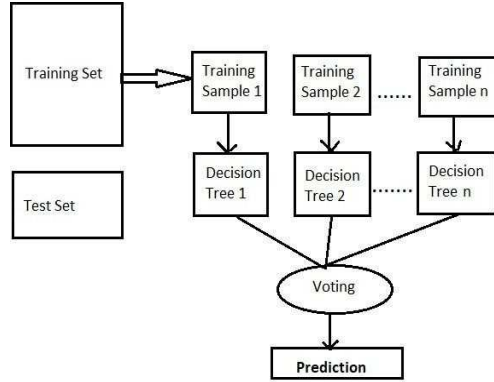


Fig.1 Working of Random Forest

Random forest is an easy-to-use machine learning method that, in most cases, produces excellent results without hyper-tuning. Overfitting is a serious issue that can sabotage results, but with the Random Forest approach, the classifier will not overfit the model if there are enough trees in the forest. The Random Forest classifier can solve the problems of missing data and can be more useful for categorical values. A random forest algorithm's building components are decision trees. A basic understanding of decision trees will aid our understanding of random forest algorithms. A decision tree mainly consists of the following components. They are

Decision node- There are two or more branches in a decision node.

Leaf node- The classification, or decision, of any branch, is represented by the leaf node. In this diagram, the diseases correspond to the leaf nodes.

Root node- Topmost decision node which classifies accurately the training data.

A random forest classifier is a combination of different decision trees like the ones shown in fig-2. Random forest is based on the wisdom of crowds, which is a simple yet powerful principle that means many trees without any relation among them are acting as a committee gives more accurate results when compared to a single prediction. The trees defend each other from their unique flaws, which results in this magnificent effect. By using the bagging method, Bagging (Bootstrap Aggregation) Trees depend greatly on the training data they are trained on, and even small changes may lead to dramatically different tree building. This leads to getting a unique prediction of disease from each decision tree.

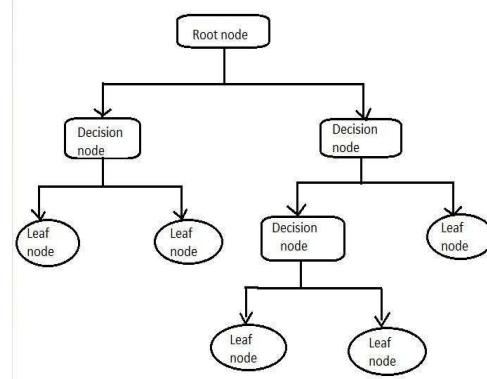


Fig.2 Decision tree, there are three types of nodes

Figure 2 describes the structure of each decision tree which the construction of a random forest algorithm. Bagging is the term for this procedure. When splitting a node in a traditional decision tree, separate the attributes as a left node, right node, and root node can be done according to the Information gain and Entropy values calculated for each attribute. But for the random forest, each tree can choose from random features with a replacement which creates more variance in the prediction by increasing the diversification of the prediction.

IV RESULTS AND DISCUSSIONS

Algorithm Performance on Training Data

The algorithm was trained using the medical records of 4920 patients who were susceptible to 41 diseases as a result of a combination of symptoms. To avoid overfitting, we took into account 95 of the 132 symptoms.

TABLE IV ACCURACY SCORE ON TESTED DATA

Algorithm used	Accuracy Score
Random Forest	0.932927

Algorithm Performance on Test Data

The system was put to the test on 41 new patient records with 95 symptoms. Below is the accuracy score as well as the confusion matrix.

GUI Result

Fig.3 A disease prediction GUI with no data.

The user enters five symptoms into the graphical user interface. When the "None" option is chosen, the user is presented with a list of symptoms to choose from. The user can list up to five symptoms he/she is experiencing. After the

symptoms have been provided, the algorithm is chosen. The symptoms are processed and the disease is searched as the algorithm is chosen.

Fig.4 Predicted result

The symptoms given by the patient ABC were: “belly pain”, “chest pain”, “hip joint pain”, “knee pain”, “back pain”.

Prediction by random forest algorithm is - Osteoarthritis

```
clean_code
C:\Python27\python.exe C:/Users/eee/Desktop/Disease-prediction/clean_code.py
C:\Python27\lib\site-packages\sklearn\ensemble\forest.py:246: FutureWarning: The
"10 in version 0.20 to 100 in 0.22.", FutureWarning: The
0.9512195121951219
39
```

Fig.5. Accuracy of random forest algorithm as 95.12%

TABLE V: EVALUATION PARAMETERS

Algorithm	Accuracy	Precision
Random Forest	95.12	0.12

Following training, the system was put to the test on 41 new patient records with 95 symptoms. Above is the accuracy score as well as the confusion matrix.

V CONCLUSION AND FUTURE ENHANCEMENT

Prevention and treatment of illness have to be identified in the early stage. Conventional methods are struggling to give accuracy and more error-prone. There will be a huge requirement for Automated Disease Prediction System that will reduce these challenges. This paper proposed a Random Forest-based classification-based algorithm for Disease Prediction System. After conducting the rigorous experiments with symptoms data set available on the internet, our proposed disease prediction model solved the challenges. 41 new records were given for testing. Our model classified 39 records correctly and only 2 records were incorrectly classified with an accuracy rate of 95% which is better than the existing system.

It can be done as a mobile application by adding more details and the latest diseases like COVID19. Every one of us wants a healthy medical system, and doctors are expected to be medical specialists who make sound decisions all of the time. However, memorizing all of the knowledge, patient history, and records required for each circumstance is exceedingly unlikely. Despite having a large amount of data and information, comparing and analyzing the symptoms of all diseases and predicting the result is tough. Predicting disease according to the previous records may lead to wrong analysis. Because health problems are cannot be estimated so continuous monitoring is necessary which is implemented in a Personalized healthcare plan, virtually by using IoMT devices.

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